

8.1.1 Independent analog peripherals supply

To improve ADC and DAC conversion accuracy and to extend the supply flexibility, the analog peripherals have an independent power supply which can be separately filtered and shielded from noise on the PCB.

- The analog peripherals voltage supply input is available on a separate V_{DDA} pin.
- An isolated supply ground connection is provided on V_{SSA} pin.

The V_{DDA} supply voltage can be different from V_{DD} . The presence of V_{DDA} must be checked before enabling any of the analog peripherals supplied by V_{DDA} (A/D converter, D/A converter, comparators, operational amplifiers, voltage reference buffer).

The V_{DDA} supply can be monitored by the peripheral voltage monitoring (PVM), and compared with two thresholds (1.65 V for PVM3 or 1.8 V for PVM4), refer to [Section 8.3.3: Peripheral voltage monitoring \(PVM\)](#) for more details.

When a single supply is used, V_{DDA} can be externally connected to V_{DD} through the external filtering circuit in order to ensure a noise-free V_{DDA} reference voltage.

ADC and DAC reference voltage

To ensure a better accuracy on low-voltage inputs and outputs, the user can connect to V_{REF+} a separate reference voltage lower than V_{DDA} . V_{REF+} is the highest voltage, represented by the full scale value, for an analog input (ADC) or output (DAC) signal.

V_{REF+} can be provided either by an external reference or by an internal buffered voltage reference (VREFBUF).

The internal voltage reference is enabled by setting the ENVR bit in the [Section 23.4.1: VREFBUF control and status register \(VREFBUF_CSR\)](#). The voltage reference is set to 2.5 V when the VRS bit is set and to 2.048 V when the VRS bit is cleared. The internal voltage reference can also provide the voltage to external components through V_{REF+} pin. Refer to the device datasheet and to [Section 23: Voltage reference buffer \(VREFBUF\)](#) for further information.

8.1.2 Independent I/O supply rail

Some I/Os from port G (PG[15:2]) are supplied from a separate supply rail. The power supply for this rail can range from 1.08 V to 3.6 V and is provided externally through the V_{DDIO2} pin. The V_{DDIO2} voltage level is completely independent from V_{DD} or V_{DDA} . The V_{DDIO2} pin is available only for some packages. Refer to the pinout diagrams or tables in the related device datasheet(s) for I/O list(s).

After reset, the I/Os supplied by V_{DDIO2} are logically and electrically isolated and therefore are not available. The isolation must be removed before using any I/O from PG[15:2], by setting the IOSV bit in the PWR_CR2 register, once the V_{DDIO2} supply is present.

The V_{DDIO2} supply is monitored by the peripheral voltage monitoring (PVM2) and compared with the internal reference voltage ($3/4 V_{REFINT}$, around 0.9V), refer to [Section 8.3.3: Peripheral voltage monitoring \(PVM\)](#) for more details.

8.1.3 Independent USB transceivers supply

The USB transceivers are supplied from a separate V_{DDUSB} power supply pin. V_{DDUSB} range is from 3.0 V to 3.6 V and is completely independent from V_{DD} or V_{DDA} .